This framework formalizes the workflow into a "Policy Extraction & Validation Protocol.”

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1. Structural Discovery: The Outline Heuristic

NDCs are not standardized; they vary by country. To find policy-rich sections, you must look for specific keywords that signal the country’s commitments.

Standard Sections

- **Mitigation:** Actions aimed at reducing greenhouse gas (GHG) emissions (e.g., transitioning to renewables).
- **Adaptation:** Actions aimed at adjusting to the current and future effects of climate change (e.g., sea walls, drought-resistant crops).
- **Means of Implementation (MoI):** Where finance, technology transfer, and capacity-building are discussed.

Non-Standard Indicators (Where to look if the above are missing)

If the outline is vague, look for these "Proxy Sections" in the Table of Contents:

- **Sectoral Planning:** Sections titled "Energy," "Transport," "Agriculture," or "Waste Management."
- **Priority Areas:** Often listed as "Key Pillars" or "Strategic Directions."
- **Cross-Cutting Issues:** Sections on gender, youth, or indigenous peoples often contain localized adaptation policies.
- **Investment Portfolios:** Lists of projects seeking international funding.

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2. Extraction: The Policy Nomenclature

Policies are rarely labeled "Policy #1." Countries use diverse terminology to describe their commitments. A comprehensive extraction tool should scan for:

- **Action-Oriented:** Action Plan, Action Item, Measure, Initiative, Intervention, Program.
- **Strategic:** Roadmap, Strategy, Long-term Vision, Policy Framework, Master Plan.
- **Regulatory:** Legislative Act, Decree, Mandate, Building Code, Standard (e.g., MEPS - Minimum Energy Performance Standards).
- **Economic:** Fiscal Instrument, Subsidy, Carbon Tax, Feed-in Tariff, Incentive Scheme.
- **Project-Based:** Pilot Project, Flagship Program, Infrastructure Investment.

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3. The Validation Framework: Valid vs. Non-Sound

The core of your system is distinguishing between "rhetoric" and "commitment." We can define a **Valid Policy** through three dimensions: **Specificity, Tangibility, and Scalability.**

A. Characteristics of a Valid Policy

A policy is valid if it contains **Observable Parameters**.

1. **Quantifiable Targets:** The policy includes numbers, dates, and units.
    - *Example:* "Retrofit 5,000 public buildings for 30% energy efficiency by 2030."
2. **Binding Mechanisms:** The policy is backed by law or regulation, not just "encouragement."
    - *Example:* "Implementing a mandatory 10% biofuel blend for all commercial transport by 2027."
3. **Spatial Specificity:** It identifies exactly *where* the impact will occur.
    - *Example:* "Restoring 500 hectares of mangroves in the [X] River Delta."
4. **Technological Shift:** It specifies a transition from one state to another.
    - *Example:* "Phasing out all coal-fired power plants with a capacity over 100MW by 2040."

B. Characteristics of a Non-Sound Policy

A policy is "non-sound" or "weak" if it is **Performative** or **Vague**.

1. **Process without Outcome:** Focuses on meetings rather than results.
    - *Example:* "Strengthening dialogue with environmental NGOs" or "Promoting awareness of climate change."
2. **Aspirational Language:** Uses verbs like "seek to," "explore," "strive," or "aim" without a roadmap.
    - *Example:* "We aim to explore the possibility of hydrogen power in the future."
3. **Lack of Baseline:** Offers a change without stating the starting point.
    - *Example:* "We will increase green spaces" (without saying by how much or where).
4. **Administrative Maintenance:** Standard government operations rebranded as climate action.
    - *Example:* "Maintaining the existing Ministry of Environment budget."

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3. District energy systems

- If framed as decarbonizing heating and cooling → **Mitigation**
- If framed as ensuring energy reliability during extreme weather → **Adaptation (Resilience)**
- Often legitimately **both**, but the dominant intent must be identified.

4. Floodable parks and wetlands

- If framed as reducing flood risk through natural absorption → **Nature-Based Solutions + Adaptation**
- If framed as biodiversity enhancement with no climate-risk framing → **Nature-Based Solutions only**

5. Infrastructure hardening

- Strengthening power grids, water systems, transport networks to withstand storms or heat → **Adaptation (Resilience)**
  Even if energy-related, this is not mitigation unless emissions reduction is the main objective.

Resilience should be a tag or attribute, not a standalone class

- Resilience is **not a distinct policy mechanism**
- It describes **how well a system performs under stress**, not what it does
- Treating it as a class leads to overlap and confusion with adaptation

Logic: Use **Adaptation** as the category, and represent resilience as a **subcategory**.

Example

- Category: Adaptation
- Sub-category: Resilience

Policy Type	Valid / High Merit	Non-Sound / Low Merit
Transport	Convert 20k city buses to electric by 2035.	Encourage citizens to use public transport more often.
Urban	Increase urban canopy cover by 2% via mandatory green roofs.	Study the benefits of trees in urban environments.
Fiscal	Implementation of a $25/ton carbon tax on heavy industry.	Support the idea of "polluter pays" in principle.
Building	Update national codes to require Net-Zero status for new builds.	Promote sustainable architecture through workshops.
Energy	Install 5GW of offshore wind capacity by 2030.	Facilitate partnerships for renewable energy research.

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5. The "Global Consistency" Filter

Valid policies are often **replicable**. A strong policy is one that has a proven "logic of intervention." If a policy like "Building Codes" appears across multiple NDCs, it gains validity because:

- **Implementation exists:** There is a known path to execute it.
- **Benchmarking is possible:** You can compare the stringency of one country's building code against another's.
- **Data availability:** These policies usually produce measurable data (e.g., energy saved per square meter).

Mitigation

Mitigation policies are designed to influence the climate system by changing the amount of greenhouse gases humans release into the atmosphere or by increasing the ability of natural or engineered systems to remove those gases from the atmosphere. The central idea behind mitigation is that human activities such as energy production, transportation, industrial processes, and land use are major contributors to climate change, and that altering these activities can slow or limit future warming.

A policy falls under mitigation when its main purpose is to reduce emissions at their source, avoid future emissions, or actively remove carbon dioxide or other greenhouse gases from the atmosphere. This includes policies that shift energy supply toward low-carbon sources, reduce reliance on fossil fuels, improve the carbon intensity of production systems, or expand carbon sinks through deliberate intervention. Even when mitigation policies deliver additional benefits, such as improved air quality or economic savings, they are classified as mitigation if their primary intent is to affect greenhouse gas concentrations and long-term climate forcing.

Mitigation policies are forward-looking in nature. They are concerned with preventing future climate change rather than responding to climate impacts that have already begun. If a policy's success is measured by emissions reduced, carbon sequestered, or progress toward decarbonization targets, it is a mitigation policy.

Operational Definition

A policy is **Mitigation** if its **primary causal pathway** is:
Human activity → ↓ GHG emissions OR ↑ carbon sinks → ↓ atmospheric forcing

What the model should look for

- Direct or indirect **emissions reduction**
- **Energy system transformation**
- **Decarbonization targets** or trajectories
- **Carbon sequestration** framed in climate terms

Typical policy mechanisms

- Renewable energy mandates
- Electrification (transport, buildings)
- Carbon pricing, offsets, caps
- Fuel switching
- Carbon capture or storage

Disambiguation rule
If the policy **acts on the climate system itself**, not on vulnerability to impacts → **Mitigation**.

Adaptation

Adaptation policies are designed to reduce the negative consequences of climate change by helping people, ecosystems, and infrastructure cope with current or expected climate impacts. Unlike mitigation, adaptation does not attempt to slow climate change itself. Instead, it assumes that climate change is already happening or will happen and focuses on reducing harm. Resilience is a core component of adaptation. It refers to a system's capacity to continue functioning under climate stress, to absorb disturbances without collapsing, and to recover quickly after disruption. A policy that strengthens resilience does so by improving robustness, flexibility, redundancy, preparedness, or recovery capacity in the face of climate hazards such as heatwaves, flooding, droughts, storms, or sea-level rise. A policy should be classified as adaptation when its primary logic begins with a climate risk and ends with reduced exposure, reduced sensitivity, or increased adaptive capacity, resulting in reduced vulnerability, reduced damage, or faster recovery. This includes actions that protect people, infrastructure, ecosystems, or services from climate impacts, even when those actions also produce economic, environmental, or emissions-related co-benefits.

If a policy measures success in terms of reduced losses, maintained service continuity, avoided disruption, or increased preparedness under climate stress, it belongs in the adaptation category. When the policy explicitly strengthens the ability to withstand or recover from climate impacts, it should be understood as resilience-focused adaptation. Adaptation policies are often place-specific and context-dependent because climate impacts vary across regions. They may involve physical interventions, institutional changes, planning processes, or social programs. Even if an adaptation policy leads to secondary benefits such as reduced emissions or economic savings, it remains classified as adaptation if its primary goal is to manage climate risk and protect people or assets from harm.

If a policy's logic begins with anticipated climate impacts and ends with increased resilience, preparedness, or safety, it should be classified as adaptation.

Operational Definition

A policy is **Adaptation** if its **primary causal pathway** is:
Climate hazard → ↓ exposure OR ↓ sensitivity OR ↑ adaptive capacity

What the model should look for

- Explicit reference to **climate impacts** (heat, flooding, drought, sea-level rise)
- **Resilience**, **preparedness**, or **risk reduction**
- Protection of people, assets, or systems from **future climate conditions**

Typical policy mechanisms

- Flood defenses
- Heat action plans
- Climate-resilient infrastructure
- Emergency response capacity
- Insurance or risk-transfer instruments

Disambiguation rule
If the policy **assumes climate change will occur and responds to its effects → Adaptation**.

Resource Efficiency

Resource efficiency policies aim to deliver the same level of service, output, or quality of life while using fewer physical resources such as energy, water, materials, or land. The core idea is that many systems are inefficient and consume more resources than necessary, and that improving efficiency can reduce costs, waste, and environmental pressure.

A policy belongs to the resource efficiency category when its primary objective is to reduce resource inputs or waste outputs per unit of activity, regardless of whether climate change is explicitly mentioned. These policies often focus on optimizing performance, improving technology, or encouraging behavioral change so that less energy, water, or material is required to achieve the same result.

Resource efficiency policies may indirectly reduce greenhouse gas emissions, especially when energy use is lowered, but emissions reduction is not the defining feature. The defining feature is improved efficiency rather than climate impact. A policy that emphasizes performance standards, efficiency benchmarks, lifecycle optimization, or waste reduction without centering its justification on climate mitigation should be classified as resource efficiency.

If a policy can be fully understood and justified even in the absence of climate change concerns, it is likely a resource efficiency policy.

Operational Definition

A policy is **Resource Efficiency** if its **primary causal pathway** is:

Same service → ↓ energy / water / material input OR ↓ waste output

What the model should look for

- Efficiency improvements **without requiring climate justification**
- Optimization of **inputs, flows, or lifecycles**
- Performance-based standards

Typical policy mechanisms

- Energy efficiency codes
- Water efficiency standards
- Circular economy policies
- Waste reduction / recycling mandates
- Industrial process optimization

Disambiguation rule

If emissions reduction is **a secondary consequence**, not the primary goal → **Resource Efficiency**.

Nature-Based Solutions

Nature-based solutions are policies that intentionally use natural systems or ecological processes to address environmental, climate, or societal challenges. The central premise is that ecosystems such as forests, wetlands, soils, and coastal systems can provide services that reduce risk, enhance resilience, or support climate goals when they are protected, restored, or managed appropriately.

A policy is classified as a nature-based solution when its effectiveness depends on the functioning of living systems rather than solely on engineered or technological infrastructure. These policies may address climate mitigation by enhancing carbon storage, adaptation by reducing exposure to hazards such as flooding or heat, or broader sustainability goals such as biodiversity protection and human well-being.

Nature-based solutions are defined by their mechanism, not their outcome. Even when a policy targets adaptation or mitigation, it should be labeled as a nature-based solution if the primary means of achieving that goal relies on ecosystems or ecological restoration. Built or hybrid systems can fall into this category when they intentionally integrate natural processes as a core component.

If removing the ecological element would fundamentally break the policy's effectiveness, the policy belongs in the nature-based solutions category.

Operational Definition

A policy is **Nature-Based Solutions** if its **primary causal pathway** is:

Ecosystem protection/restoration → climate OR resilience OR co-benefits

What the model should look for

- Explicit use of **ecosystems as infrastructure**
- Restoration, conservation, or enhancement of natural systems
- Solutions that rely on **biological processes**

Typical policy mechanisms

- Urban forests
- Wetland restoration
- Green roofs and corridors
- Mangroves, dunes, riparian buffers

Disambiguation rule

If the **mechanism depends on living systems** rather than engineered systems → **NbS**.

Cross-Category Resolution Rules

When classifying policies, priority should always be given to the policy's primary causal logic rather than secondary benefits, aspirational language, or co-benefits. A single policy may legitimately belong to more than one category, but each category should only be assigned when the underlying cause–effect relationship clearly matches the definition described above.

These rules prevent confusion and improve classification accuracy.

Rule 1: Primary mechanism > secondary benefits
Classify by **what the policy does**, not by claimed co-benefits.

Example:
"Energy efficiency to reduce emissions" → **Resource Efficiency**, not Mitigation.

Rule 2: Climate system vs climate impacts

- Acts on **emissions/carbon cycle → Mitigation**
- Acts on **hazards/vulnerability → Adaptation**

Rule 3: Natural vs engineered pathways

- Ecosystem-mediated → **NbS**
- Technological / infrastructural → other categories

Rule 4: Multi-label is allowed
Policies can legitimately be:

- Mitigation + NbS
- Adaptation + NbS
- Resource Efficiency + Mitigation, but the **primary label should be identifiable**.

Edge Cases for Adaptation Classification

1. Green roofs

- If framed as reducing urban heat exposure, managing stormwater, or protecting buildings from extreme weather → **Adaptation (Resilience)**
- If framed primarily as ecosystem restoration or urban ecology → **Nature-Based Solutions**
- If framed primarily as carbon storage → **Mitigation (secondary)**

Primary mechanism decides.

2. Passive cooling and shading

- If framed as protecting occupants during heatwaves or maintaining livable indoor temperatures during power outages → **Adaptation (Resilience)**
- If framed as reducing energy demand or emissions → **Resource Efficiency**

Classification Logic: is the goal survival and comfort under heat stress, or reduced energy consumption?

Secondary Category Typology

The twelve types are grouped below by primary category.

Mitigation primary

M-1 Efficiency-enabled Mitigation (secondary: Resource Efficiency)
M-2 Resilience-integrated Mitigation (secondary: Adaptation)
M-3 Sequestration-based Mitigation (secondary: Nature-Based Solutions)

Resource Efficiency primary

RE-1 Emissions-quantified Efficiency (secondary: Mitigation)
RE-2 Climate-resilient Efficiency (secondary: Adaptation)
RE-3 Ecosystem-supported Efficiency (secondary: Nature-Based Solutions)

Nature-Based Solutions primary

NBS-1 Hazard-responsive NBS (secondary: Adaptation)
NBS-2 Sequestration-delivering NBS (secondary: Mitigation)
NBS-3 Resource-saving NBS (secondary: Resource Efficiency)

Adaptation primary

A-1 Ecosystem-mediated Adaptation (secondary: Nature-Based Solutions)
A-2 Efficiency-supported Adaptation (secondary: Resource Efficiency)
A-3 Emissions-coupled Adaptation (secondary: Mitigation)

Primary Category: Mitigation

The policy's primary mechanism directly reduces greenhouse gas emissions at the source. The causal chain runs from human activity to atmospheric forcing: human activity leads to reduced GHG emissions, which leads to reduced atmospheric forcing.

Type M-1: Efficiency-enabled Mitigation

Secondary category added: Resource Efficiency

Definition.

The policy achieves its emissions reduction goal partly through an efficiency mechanism, reducing resource inputs per unit of service delivered. Efficiency is not a side-effect but a named part of the emissions strategy.

Causal chain.

Human activity leads to reduced GHG emissions via a named efficiency action combined with fuel switching or decarbonization.

Classification rules.

- The policy text names both an emissions target and a specific efficiency action. Qualifying efficiency actions include operational improvements, route optimization, right-sizing, building performance standards, and procurement efficiency requirements.
- The efficiency action is framed as a delivery mechanism for the emissions goal, not as a separate program.

- The outcome is primarily measured in emissions units. Efficiency is the instrument used to achieve it.
- Do not apply if the only efficiency language is general, such as "improve performance", with no named action.

Example.
"The City will reduce municipal fleet emissions by 50% by 2025 through electrification, cleaner fuels, efficiency, and procurement standards as guided by the Green Fleet Action Plan."
The commitment names efficiency as a mechanism alongside the emissions target. The source quote elaborates: "increasing efficiency in how vehicles are used", naming an operational efficiency action. Both the commitment and source quote are valid evidence. Resource Efficiency is added as a secondary category.

Type M-2: Resilience-integrated Mitigation

Secondary category added: Adaptation

Definition.
The policy contains both an emissions reduction goal and a named climate resilience or emergency preparedness requirement. Both mechanisms are present in the same commitment.
Causal chain.
Human activity leads to reduced GHG emissions, and within the same policy, a climate hazard is addressed through a mandated resilience measure.
Classification rules.

- A specific climate hazard or resilience need is named in the same policy commitment. Qualifying hazards include heat, flood, drought, storm surge, and sea-level rise.
- A resilience or emergency response measure is required, not just acknowledged as a co-benefit.
- The adaptation component must be a mandated action, not a general statement about building climate resilience.
- Energy storage policies that explicitly state both a decarbonization function and a grid resilience function under climate stress qualify, provided the resilience function is named in the policy text or source quote, not only inferred by the reviewer.
- Do not apply if resilience is mentioned only in reviewer notes or background context, without appearing in the policy commitment or source quote itself.

Type M-3: Sequestration-based Mitigation

Secondary category added: Nature-Based Solutions

Definition.
The policy uses biological carbon sequestration in an urban ecosystem, such as tree canopy, wetlands, or urban forests, as the mechanism for achieving its carbon storage goal.
Causal chain.
An ecological system performs biological carbon uptake, contributing to a net GHG reduction.

Classification rules.

- The ecological action, such as tree planting, afforestation, or wetland restoration, is explicitly named as the mechanism for achieving the emissions or sequestration goal.
- A carbon storage, sequestration, or emissions offset target is stated in the policy text, not inferred.
- The ecological action is deliberate and specific, not incidental green space.
- Do not apply simply because trees absorb carbon as a biological property. The policy must name sequestration as a goal.
- This type is the inverse of NBS-2. Here the policy is Mitigation primary and uses ecology as its sequestration instrument. In NBS-2, the policy is NBS primary and sequestration is a named secondary outcome.

Dataset note: No cases of this type are present in the current 86-policy dataset. All policies in the dataset that involve tree planting or afforestation were classified as Nature-Based Solutions primary, not Mitigation primary. Apply this type when a policy is structured around an emissions target that is delivered through a named biological sequestration mechanism.

Primary Category: Resource Efficiency

The policy reduces resource inputs, including energy, water, and materials, per unit of service delivered. GHG reductions are a co-benefit, not the primary mechanism. The causal chain runs from same service to reduced resource input to an implied co-benefit of reduced GHG emissions.

Type RE-1: Emissions-quantified Efficiency

Secondary category added: Mitigation

Definition.

The efficiency policy also states an explicit GHG or emissions reduction target, a number, percentage, or named commitment, making Mitigation a second distinct mechanism alongside the efficiency goal.

Causal chain.

Same service leads to reduced resource input, and a named GHG target is also stated in the same policy commitment.

Classification rules.

- An explicit emissions target must appear in the policy text or source quote. Qualifying forms include a percentage GHG or carbon reduction, a ton figure, or a named goal such as carbon neutral or net zero.
- The target must be tied to this policy commitment, not to a separate city-wide goal mentioned only in background context.
- The expectation that efficiency reduces emissions is not sufficient. The target must be stated. This is the critical test distinguishing RE-1 from a plain Resource Efficiency classification.
- For indirect instruments such as benchmarking or disclosure-only policies, do not apply this type unless the disclosure policy also states an explicit GHG target. Flag these records as indirect instruments.

Example.
"Double the budget allocation for municipal building energy reductions (2021–2025) and target a 40% reduction in energy use and carbon emissions in municipal buildings by 2025."
The policy names both an energy efficiency target and an explicit carbon emissions reduction target. Both mechanisms are present. Mitigation is added as a secondary category.

Counter-example.
"Require commercial buildings 50,000 sq ft or larger to complete Building Tune-Ups (operations & maintenance improvements) under SMC 22.930, with phased compliance beginning in early 2019."
The commitment names an operational efficiency action, mandatory tune-ups, with no GHG target stated anywhere in the policy text or source quote. Mitigation is not added as a secondary category, even though building energy reductions will reduce emissions as a co-benefit.

Type RE-2: Climate-resilient Efficiency

Secondary category added: Adaptation

Definition.
The efficiency measure is introduced specifically in response to a named climate-driven scarcity or hazard. Examples include a water conservation mandate prompted by projected drought, or energy demand management for heat-driven grid stress.

Causal chain.
A climate hazard creates resource scarcity, which motivates an efficiency mandate, which reduces resource demand and reduces vulnerability to the hazard simultaneously.
Classification rules.

- The policy text must name the climate hazard or scarcity condition that motivates the efficiency requirement.
- The efficiency measure must be framed as a response to the hazard, not as a general efficiency improvement that happens to improve resilience.
- Qualifying language includes phrases such as "in response to drought conditions", "to manage heat-driven peak demand", or "to reduce water scarcity risk".
- Do not apply if the climate connection is only implied or appears in background context rather than in the policy commitment or source quote.

Type RE-3: Ecosystem-supported Efficiency

Secondary category added: Nature-Based Solutions

Definition.
Natural systems are used as a mechanism to deliver resource savings. Examples include green roofs reducing building cooling load, urban tree canopy reducing energy demand, or bioswales reducing treated water demand.
Causal chain.
An ecological mechanism reduces resource demand per unit of service, for example less cooling energy or less stormwater treatment.

Classification rules.

- A specific ecological intervention must be named as the mechanism that achieves the resource savings.
- The resource saving, whether energy or water, must be the primary stated goal of the ecological action, not the ecological goal itself.
- This type is rare. Most NBS policies have ecology as the primary goal. Apply this type only when the resource efficiency outcome is primary and the ecological mechanism is explicitly named as its instrument.
- Do not apply when the NBS program is listed separately from the efficiency targets.

Primary Category: Nature-Based Solutions

The policy uses ecosystems or natural processes as the primary mechanism. The causal chain runs from an ecological mechanism to urban co-benefits including cooling, flood control, carbon sequestration, and biodiversity.

Type NBS-1: Hazard-responsive NBS

Secondary category added: Adaptation

Definition.

The ecological intervention addresses a named climate hazard, such as heat island, stormwater flooding, coastal erosion, or desertification, alongside its primary ecological goals. The NBS is simultaneously doing adaptation work.

Causal chain.

An ecological mechanism reduces exposure to a named climate hazard while also delivering broader ecosystem services.

Classification rules.

- A specific climate hazard must be named in the policy text or source quote. Qualifying hazards include heat island, flood, drought, storm surge, desertification, and dust storms.
- The ecological action must be the mechanism through which the hazard is addressed, not a separate engineered measure.
- The NBS remains primary because the ecological system is the intervention. The hazard provides a second framing for what the ecology is delivering.
- This is the most common NBS secondary type. Apply it when heat reduction, stormwater management, or desertification control appear as explicit stated goals.
- Where a climate hazard is identified only in the causal mechanism analysis by the classifier and not stated in the policy text itself, flag this assignment in the classification notes as inferred rather than explicit. Do not treat it as a clean NBS-1 assignment.

Example.

"Implement national green-belt / ‘Green Wall’ afforestation projects (e.g., plant 315,000 trees to form a 420 km wall and establish green belts of 50–180 km length and 5–20 km width) to reduce dust and adapt to desertification."

This policy is classified as Nature-Based Solutions primary with Adaptation secondary in the dataset. The ecological system (afforestation) is the intervention, making NBS the primary category. The hazards are stated explicitly in the text (dust reduction, desertification adaptation), triggering NBS-1.

Type NBS-2: Sequestration-delivering NBS

Secondary category added: Mitigation

Definition.
The ecological intervention explicitly delivers carbon sequestration or contributes to an emissions reduction target. The NBS is acting as a carbon sink alongside its primary ecological function.
Causal chain.
An ecological mechanism produces biological carbon uptake, contributing to a GHG reduction goal.
Classification rules.

- Carbon sequestration, carbon storage, or an emissions offset must be named as a stated goal in the policy text or source quote.
- The ecological action must be identified as the sequestration mechanism.
- Do not apply simply because trees absorb carbon as a biological property. The policy must state this as an intended outcome.
- A specific sequestration target or explicit link to a city-wide carbon goal strengthens the case for this type.
- This type often co-occurs with NBS-1. A canopy policy may address both heat island reduction and carbon storage simultaneously.

Example.
"Legally protect an additional 20,000 acres of natural carbon pools and bring total managed natural areas to ~70,000 acres by 2030, prioritizing resilience and benefits for low-income communities and communities of color."
The phrase "natural carbon pools" appears in both the commitment and source quote, explicitly naming carbon storage as a named property and goal of the ecological action. Mitigation is added as a secondary category. This is the only case in the current dataset where NBS-2 can be applied on fully explicit grounds.

Type NBS-3: Resource-saving NBS

Secondary category added: Resource Efficiency

Definition.
The ecological intervention is designed to reduce resource consumption. Examples include green infrastructure that reduces stormwater treatment demand, or urban tree canopy that reduces building cooling loads and energy use.
Causal chain.
An ecological mechanism reduces the resource input required, whether energy, water, or treatment capacity.
Classification rules.

- A specific resource saving, whether energy, water, or stormwater treatment capacity, must be named as a stated goal.
- The ecological mechanism must be identified as the cause of the resource saving.
- This type is rare. Apply only when resource savings are a primary stated goal, not an implied property of green infrastructure.
- Do not apply when the ecological goal, such as biodiversity or greenspace, is primary and resource savings are not mentioned.

Primary Category: Adaptation

The policy reduces vulnerability to climate impacts that are already occurring or projected. The primary logic is risk management. The causal chain runs from climate hazard exposure to reduced vulnerability to avoided damage.

Type A-1: Ecosystem-mediated Adaptation

Secondary category added: Nature-Based Solutions

Definition.
The adaptation measure uses an ecological mechanism, such as wetland restoration, afforestation, or green belts, as the chosen instrument to reduce exposure to the named climate hazard. Nature is the delivery vehicle for the adaptation outcome.
Causal chain.
A climate hazard is addressed through an ecological mechanism, which reduces vulnerability or hazard exposure.
Classification rules.

- An ecological action must be explicitly named as the mechanism of adaptation. Qualifying actions include tree planting, wetland restoration, afforestation, bioswales, and green belts.
- The ecological action must be positioned as the means of reducing the climate hazard, not as a separate biodiversity or greenspace goal.
- The adaptation goal remains primary because the policy exists to address the named hazard. Ecology is the instrument chosen to do so.
- If the policy also mandates engineered measures alongside the ecological ones, A-1 still applies to the ecological component.
- Do not apply if the ecological action is described in general terms without being named as the mechanism of hazard reduction.

Type A-2: Efficiency-supported Adaptation

Secondary category added: Resource Efficiency

Definition.
The adaptation plan includes explicit resource efficiency targets tied to the climate vulnerability being addressed. Examples include water use efficiency benchmarks embedded in a drought adaptation programme, or energy demand management within a heat resilience plan.
Causal chain.

A climate hazard motivates an adaptation plan that includes a named efficiency measure as one of its components, reducing resource demand and vulnerability simultaneously.

Classification rules.

- A direct efficiency action or target must be stated in the policy commitment or source quote, not just a general intention to use resources wisely.
- The efficiency measure must be named as a component of the adaptation plan, not as a separate program.
- Qualifying examples include water use efficiency targets in a drought adaptation program, and energy performance standards in a heat resilience infrastructure plan.
- Monitoring indicators and sector targets that include efficiency metrics also qualify, for example water use efficiency in agriculture as part of a national adaptation program.
- Do not apply if the efficiency connection is only implied by the nature of the adaptation measure rather than stated explicitly.

Type A-3: Emissions-coupled Adaptation

Secondary category added: Mitigation

Definition.

The adaptation investment explicitly includes an emissions reduction component. Examples include low-carbon materials specified for resilience infrastructure, fuel switching in emergency systems, or a funding mechanism that combines resilience and mitigation projects.

Causal chain.

A climate hazard motivates an adaptation investment that also includes a named emissions reduction component, reducing vulnerability and reducing GHG simultaneously.

Classification rules.

- An explicit emissions reduction or decarbonization component must be named in the same policy commitment or source quote.
- The word mitigation appearing in a funding or policy title is sufficient if it refers to GHG mitigation rather than disaster risk mitigation. Confirm from context.
- Qualifying examples include a resilience fund that explicitly covers both climate adaptation and GHG reduction projects, and emergency infrastructure built to both climate-proof and low-carbon specifications.
- Do not apply when "mitigation" appears only as disaster risk mitigation language, meaning reducing the severity of climate impacts, rather than GHG emissions mitigation. These two uses of the word must be distinguished carefully.
- This type is rare. Most adaptation policies do not include an explicit emissions component.

Common Errors to Avoid

The following mistakes account for most misclassifications in secondary category assignment.

Error 1. Adding Mitigation as secondary to any Resource Efficiency policy because efficiency reduces emissions.

Type RE-1 applies only when an explicit GHG target is stated in the policy text or source quote. The implied co-benefit is not sufficient. This is the most frequent error in the current dataset, affecting approximately 20 of the 23 RE to Mitigation assignments.

Error 2. Adding NBS secondary to Mitigation because a city has a tree planting program mentioned elsewhere.
Type M-3 requires that biological sequestration is the explicit emissions mechanism named in this policy commitment. Context from other policies or a city's general strategy does not count.

Error 3. Adding Adaptation secondary to every NBS policy because climate change is the reason cities plant trees.
Type NBS-1 requires a specific hazard to be named in this policy text. A general reference to climate change is not a named hazard. Where the hazard is identified through causal analysis but not stated in the text, flag the assignment as inferred.

Error 4. Confusing disaster risk mitigation with GHG mitigation when applying Type A-3.
Confirm from context that the word mitigation refers to greenhouse gas emissions. If it refers to disaster risk management, reducing the severity of hazard impacts, it remains Adaptation language and Mitigation should not be added as a secondary category.

Error 5. Assigning Type RE-2 because efficiency generally helps resilience.
The climate hazard must be named as the specific motivation for this efficiency requirement, not implied by the general logic that efficient systems are more resilient.

Error 6. Assigning Types RE-3, NBS-3, or M-3 on the basis of plausible mechanisms.
These three types have no confirmed cases in the current dataset. They are defined here for completeness. Apply them only when the secondary mechanism is explicitly stated as the purpose of the primary action, not because the mechanism is plausible or logical given the policy context.